FORWARDS, FUTURES, AND SWAPS

Learning Outcome

- define forward contracts, futures contracts, swaps, options (calls and puts), and credit derivatives and compare their basic characteristics

Forwards, futures, and swaps are the most common derivative contracts which represent a firm commitment. This firm commitment is an obligation of both counterparties to perform under the terms of the derivative contract. Key common features of this type of derivative include the following:

- A specific contract size
- A specific underlying
- One or more exchanges of cash flows or underlying on a specific future date or dates
- Exchange(s) based on a pre-agreed price

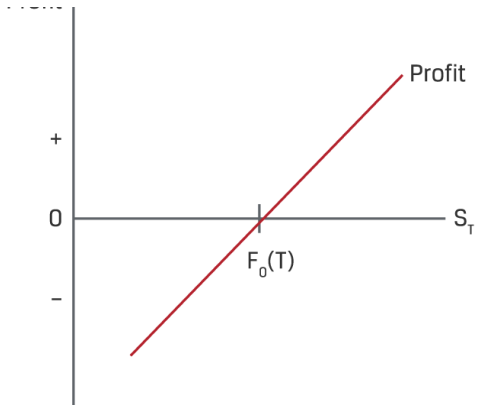
Despite their similarities, forwards, futures, and swaps each have different features, which are the subject of this lesson.

A forward contract is an over-the-counter (OTC) derivative in which two counterparties agree that one counterparty, the buyer, will purchase an underlying from the other counterparty, the seller, in the future at a pre-agreed fixed price. As noted earlier, OTC derivatives offer greater flexibility and customization than exchange-traded derivatives (ETD), but also usually involve more counterparty risk. Forward contracts are advantageous for derivative end users seeking to hedge an existing or forecasted underlying exposure based on specific terms. For example, an importer may enter a forward contract to buy the foreign currency needed to satisfy the commercial terms of a future goods delivery contract. Forward contracts are more flexible as to the size, underlying details, maturity, and/or credit terms than a similar ETD. A forward contract buyer has a long position and will therefore benefit from price appreciation of the underlying over the life of the contract.

To gain a better understanding of forwards, we must examine their payoff profile. Assume a forward contract is agreed at time $t = 0$ and matures at time T. At time $t = 0$, the counterparties do not exchange a payment upfront but, rather, agree on delivery of the underlying at time T for a **forward price** of $F_0(T)$. The subscript refers to the date on which the underlying price in the future is set ($t = 0$), and the T in parentheses refers to the date of exchange ($t = T$). The spot price of the underlying at time T is S_T. Exhibit 1 shows the payoff from the forward buyer's perspective, which is a long forward position. Note that the payoff equals the profit, as no upfront payment is made.

Exhibit 1:

Long Forward (Forward Buyer) Payoff Profile



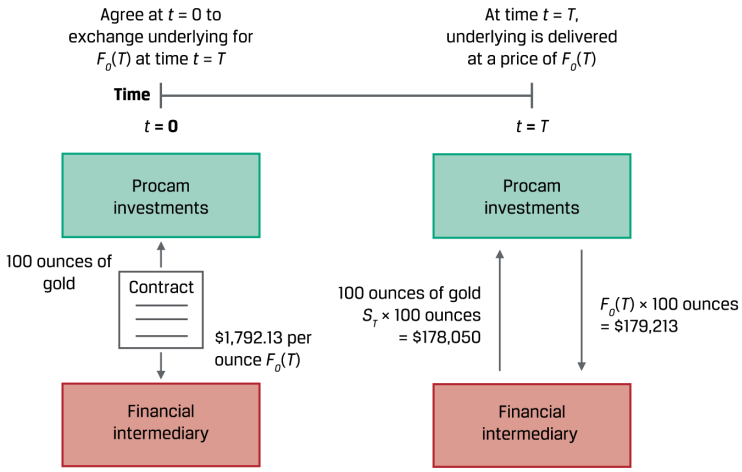
Outcome	Buyer Payoff	Seller Payoff
$S_T > F_0(T)$	$[S_T - F_0(T)] > 0$	$[F_0(T) - S_T] < 0$
$S_T < F_0(T)$	$[S_T - F_0(T)] < 0$	$[F_0(T) - S_T] > 0$

The symmetric payoff profile shown in Exhibit 1 is a common feature of firm commitments. Since the derivative price is a linear function of the underlying, firm commitments are also referred to as **linear derivatives**. At time T, the transaction is settled based on the difference between the forward price, $F_0(T)$, and the underlying price of S_T, or $[S_T - F_0(T)]$ from the buyer's perspective. That is, the buyer realizes a gain if she is able to take delivery of the underlying at a market value, S_T, that exceeds the pre-agreed price, $F_0(T)$. If the forward price exceeds the current market value $[F_0(T) > S_T]$, the buyer realizes a loss and must either take delivery of an asset at a loss of $[F_0(T) - S_T]$ or pay the seller this amount in cash. Forward contracts usually involve a single exchange in the future, as in Example 1.

EXAMPLE 1

An investor, Procam Investments, enters a cash-settled forward contract with a financial intermediary to buy 100 ounces of gold at a forward price, $F_0(T)$, of $1,792.13 per ounce in three months.

1. Today's spot gold price (S_0) is $1,770 per ounce.
2. At contract maturity, the gold price (S_T) is $1,780.50 per ounce.
3. The payoff, $S_T - F_0(T)$, is –$11.63 = $1,780.50 – $1,792.13 per ounce.
4. Procam (the buyer) must pay the financial intermediary (the seller) $1,163 (= 100 × $11.63) to settle the forward contract at maturity.



The contract may specify either the actual delivery of the underlying or a cash settlement. The settlement amount is equal to $[S_T - F_0(T)]$ from a buyer's perspective and $-[S_T -$

long forward position in the cash market.

Question Set: Forwards

1. Describe a scenario in which a forward contract has cash settlement of zero at maturity and neither counterparty has defaulted.

Hide Solution

Solution:

A forward contract will have a cash settlement of zero at maturity if $S_T = F_0(T)$ or the payoff from the buyer's perspective is $[S_T - F_0(T)] = 0$. This is often referred to as the **breakeven point** for the forward contract for both buyer and seller in the absence of transaction costs and is visually represented by the *x*-axis intercept of the profit line in Exhibit 1.

2. Determine the correct answers to fill in the blanks: An oil producer enters a derivative contract with an investor to sell 1,000 barrels of oil in two months at a forward price of $64 per barrel. If the spot oil price at maturity is $58.50 per barrel, the investor realizes a _____ at maturity equal to _____.

Hide Solution

Solution:

An oil producer enters a derivative contract with an investor to sell 1,000 barrels of oil in two months at a forward price of $64 per barrel. If the spot oil price at maturity is $58.50 per barrel, the investor realizes a *loss* at maturity equal to *$5,500*.

The oil forward price, $F_0(T)$, under the contract equals $64 per barrel.

At contract maturity, the spot oil price (S_T) is $58.50 per barrel.

• Investor payoff per barrel:

$[S_T - F_0(T)] = \$58.50 - \$64.00 = -\$5.50$ per barrel.

• Total amount the investor pays the oil producer to settle the forward contract for 1,000 barrels at maturity:

1,000 × $5.50 = $5,500.

3. Identify the most likely forward contract participants that correspond to the following statements:

A. forward contract purchaser	1. Seeks to benefit from underlying price depreciation
B. Forward contract seller	2. Realizes a gain if the initial spot price of the underlying, S_0, exceelds the forward price of $F_0(T)$
C. Neither a forward contract purchaser nor a seller	3. Receives a positive payoff at maturity if the spot price, S_T, exceeds the forward price of $F_0(T)$

Hide Solution

Solution:

1. The correct answer is B. A forward seller pays $[F_0(T) - S_T]$ to the forward contract buyer at maturity and therefore benefits as the underlying spot price S_T declines over time.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

settlement is based on the future spot price, S_T.

3. The correct answer is A. The forward contract buyer realizes a gain at maturity if $S_T > F_0(T)$.

Discuss Filter

Discussion

What do you want to discuss... (required)

ML Exceelds

Created 6 months ago by Maximos Leopoldo von Hafner 3 replies | Last Activity: 3 months ago

Show All replies Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search